Solar Home Lighting

Jack Holdridge
Kyle Walcher
Matthew Greer
Adam Garcia

Concept of Operations

Concept of Operations
for
Solar Home Lighting

Team 3

Approved by:

Jack Holdridge　　　　Date

Prof. Kalafatis　　　　Date

Hailey Ward　　　　Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	09/17/25	Jack Holdridge		Draft Release

Table of Contents

Table of Contents........3
List of Tables........5
List of Figures........6
1. Executive Summary........6
2. Introduction........7
2.1. Background........7
2.2. Overview........8
2.3. Referenced Documents and Standards........9
3. Operating Concept........10
3.1. Scope........10
3.2. Operational Description and Constraints........10
3.3. System Description........10
3.4. Modes of Operations........11
3.5. Users........11
3.6. Support........11
4. Scenarios........12
4.1. Airbnb Host #1........12
4.2. Independent/Off-Grid #2........12
4.3. Nighttime Security Monitoring........12
5. Analysis........12
5.1. Summary of Proposed Improvements........12
5.2. Disadvantages and Limitations........13
5.3. Alternatives........13
5.4. Impact........14

List of Tables

No table of figures entries found.

List of Figures

Figure 1. Solar Panel Security System Block Diagram..1

1. Executive Summary

This project aims to develop a smart solar-powered lighting system for both indoor and outdoor residential use. The system will integrate a photovoltaic system, battery storage, motion detection, and user interface to operate lights. Outdoor lighting will activate only at night when human activity is detected via motion sensors, and a camera security system will be active during the day. Both indoor and outdoor lighting will be manually controlled via a smart switch, and wirelessly controlled and monitored through a mobile application to adjust lighting preferences and schedules accordingly. This will present interdisciplinary collaboration across hardware, embedded systems, and software development, while prioritizing modular design, real-time responsiveness, and user-centered functionality..

2. Introduction

This project proposes the design and implementation of a solar-powered residential home lighting and security system with a built-in camera. The system addresses energy costs, demand for renewable power solutions, and affordable, reliable home security. The system also features an application that gives live feedback on battery, security alerts, and switches to turn lights/camera off and on remotely.

2.1. Background

Standard and conventional home lighting/security systems typically use grid power to implement their products. While effective, these systems often increase household energy and have a potential for power outages. This can leave homes unprotected when reliability is most crucial. These systems also require high energy consumption, leading to an increase in cost to maintain within a residential home. Commercial smart security systems provide live video and mobile notifications, but rely on subscription-based cloud services and stable electrical infrastructure as well. Meanwhile, solar-powered lighting systems exist, but they are usually limited to basic outdoor applications without intelligent applications and integration into a complex system such as a home.

Recent advances in low-cost microcontrollers and embedded AI/Machine Learning make it possible to design systems that use renewable energy with security at a fraction of the cost of other commercial products. The ESP32 and its built-in wifi capabilities enable direct communication between devices for real-time status updates and controls. Integrating efficient controls, such as a motion sensor to trigger the camera for security alerts, demonstrates the project's sustainable alternative.

According to the Federal Bureau of Investigation, the average dollar loss per burglary offense for a home was over $97,000 in 2022. This alarming number would devastate and financially ruin families. Not only financially, but potential burglary and home invasion could harm residents' safety. According to a North Carolina study, around 60% of surveyors would be deterred by an alarm and potentially seek another home. In the modern day and age, it is essential and a necessity to have intelligent lights and security systems. Our system seeks to address this issue at a budget-friendly price.

2.2. Overview

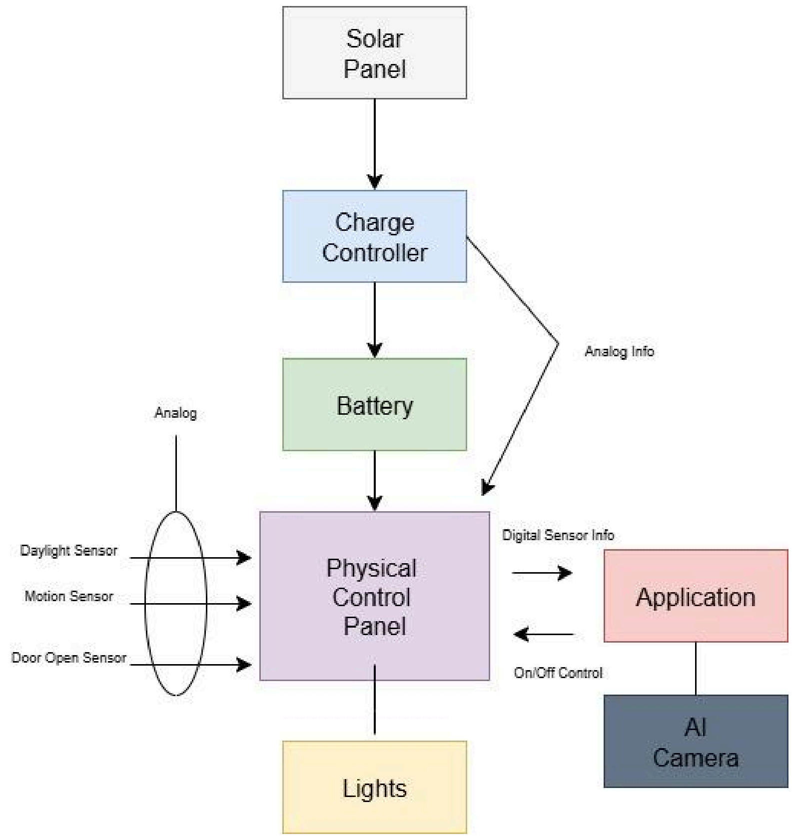


Figure 1: Solar Panel Security System Block Diagram

Our system integrates renewable energy and camera security systems by using a solar panel and charge controller to supply the solar power to a battery. This then works by powering a physical control panel built around multiple ESP32 microcontrollers. The control panel will then take in analog signals from daylight, motion, and door opening sensors. In contrast, an ESP32 and motion sensors will also handle person detection, allowing the camera to capture human presence and send security notification signals to the mobile application. In the application, the user will be able to receive camera gallery, battery percentage, light switch controls, and security notifications. Conclusively, this design provides a robust, low-cost, energy-efficient, and user-friendly solution for a security and light system in a residential home.

2.3. Referenced Documents and Standards

[1] IEEE Standards Association, IEEE Standard for Information Technology—Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks—Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications, IEEE Std 802.11™, 2020. Accessed at: https://standards.ieee.org/standard/802_11-2020.html

[2] Apple Inc., Human Interface Guidelines for iOS, Apple Developer Documentation, 2023.Accessed at: https://developer.apple.com/design/human-interface-guidelines/ios/overview/introduction/

[3] International Electrotechnical Commission (IEC), IEC 61724: Photovoltaic system Performance monitoring - Guidelines for measurement, data exchange and analysis, IEC Standard, 2017. Accessed at: https://webstore.iec.ch/publication/33622

3. Operating Concept

3.1. Scope

Design, build, and implement a smart solar-powered lighting and security system for both the interior and exterior of residential homes. The system will operate using renewable solar battery banks. This will be operated by both a local physical control panel as well as a remote app. For demonstration purposes, a functional model of a porch and fourier mounted system will be built, but the final design will be expandable to multiple interior rooms and exterior areas. This system will combine the functionality of commonly available lighting and separate smart security systems into one cohesive solar-powered unit.

3.2. Operational Description and Constraints

The system is intended to provide a modular lighting and security camera network for residential homes that is functioning entirely on solar power, independent from the power grid. The system will be controlled off of a local physical interface as well as a Wi-Fi-based app. The lighting aspect of the system will function with a dual purpose of recreational lighting for the interior and exterior of the home, as well as exterior security flood lights. The security camera portion will operate within the app to notify the user of detected motion outside of the home. There will be several motion sensors outside the house to detect possible trespassers and trigger both the camera functionality and security lights. For recreational lighting purposes, there will be motion sensors, door sensors, and a programmable timer to automatically engage or disengage the lights.

The following are constraints relating to the operational description:

- The solar panels need to be placed in an area that provides consistent direct sunlight during the day.
- The battery bank needs to be sized according to the number of lights to ensure adequate operating time.
- The physical control panel needs to be mounted within the range of the home's wifi to allow app functionality.
- Users are required to have a smartphone to use the security feature of this system.

3.3. System Description

The smart solar home lighting system integrates a renewable energy source, control electronics, and user-friendly software to provide a reliable and efficient home lighting and security system. The system consists of 4 subsystems: the solar panel network, microcontroller, sensors and lighting, and the application. This interconnected design ensures a sustainable and user-accessible solution for a modern home.

- Solar Panel Network: The solar panel array will be a roof-mounted unit fed through a charge controller into deep-cycle batteries. The batteries will collect a charge throughout the day while relaying the charge level to the user. The battery system will be designed to hold a multi-day charge to account for inclement weather interfering with charging conditions.

- Microcontroller: The microcontroller will interface with the mobile app, sensors, solar/battery network, and lighting. It reads inputs and applies the logic accordingly. It also manages comms, low power modes, enforcing limits when needed.
- Sensors and lighting: Motion sensors and cameras will communicate with the MCU when motion is detected. Once sensors are triggered, home lighting for designated areas will turn on, and users will be notified through the mobile app. Areas with cameras will turn on once motion is detected.
- Application: The Application will act as a tool that stores important system data (Camera, battery percentage, light status, and database logs) while also acting as a wireless control panel for basic operation, such as turning lights on and off. The application will communicate through Wifi via IOS device and an ESP32. The ESP32 will be mounted and connected to our main control panel, where signals will be able to receive and send out vital data.

3.4. Modes of Operations

The lighting system will function in three modes: automatic, manual, and security.

The automatic controls will function based on the state of motion, light, and door sensors, as well as user-determined automatic timeouts. The manual mode will be the operation of on and off switches, both locally with panel-mounted switches, and the app-controlled inputs.

The security mode of the system will function on the exterior of the home, where motion will trigger the camera detection and flood lights to turn on.

3.5. Users

The smart solar home lighting system is intended to be used by any homeowner. The system is simple and user-friendly, requiring minimal technical training for standard operation. To interact with the system, the users will be provided with an in-app manual that walks them through the use of the system.

For installation, basic home improvement skills are needed to mount the solar panels and route cables to the lights. The design will contain modular sensors and light plugs that allow for ease of installation.

This setup will benefit typical homeowners who want to lower their electricity bill and desire a more vigilant mindset on home security. Additionally, this system will benefit homes in more remote areas with residences separate from the local power grid.

3.6. Support

A user manual will be provided with the system that gives details on use, installation, and troubleshooting. Additionally, the app will have a walkthrough mode available to teach users the process for controlling and monitoring the system within the application.

4. Scenarios

4.1. Airbnb Host #1

- Homeowners use the app to schedule porch lights to turn on at 7 PM for expected visitors.
- The system overrides motion detection during scheduled hours.

4.2. Independent/Off-Grid #2

- The system operates independently of grid power, ideal for cabins or remote homes.
- Reliable lighting and security without utility infrastructure.

4.3. Nighttime Security Monitoring

- The system is utilized when outside monitoring is needed for safety reasons
- Smart lighting and motion detectors can store camera data in the app in a safe location.
- Motion-detected lights can deter intruders.

5. Analysis

5.1. Summary of Proposed Improvements

The proposed solar home lighting system offers several important advantages over traditional home lighting systems, such as:

- Eco-friendly/ Reduced Emissions: By utilizing solar energy as the primary power source, the system reduces reliance on natural gas and lowers gas emissions.
- Off-Grid Back-Up Use: The integration of battery storage allows the lighting system to function independently of the electrical grid, ensuring reliable operations during outages.
- Energy Efficiency: The use of LED lighting and power management works to minimize energy consumption while handling high output, contributing to extending the life of the system.
- Increased Security: Motion sensors and app-based controls enhance home security by providing responsive lighting to movement in the area.
- Mobile App Control: The System can be controlled remotely via smartphone app that offers users convenience to monitor energy use, set timers, and adjust lighting preferences from any location.
- Scalability: The modular design allows for the expansion of a porch or foyer lighting to cover both interior and exterior household lighting, adapting to changing user needs.
- Lower Energy Cost: By reducing dependence on grid electricity, households experience lower monthly utility costs.

5.2. Disadvantages and Limitations

While the proposed solar home lighting system provides several advantages, there are some limitations that need to be considered, which include:

- High Upfront Cost: The initial investment for solar panels, batteries, and smart controls is higher than conventional lighting systems. Long-term energy settings will offset the cost, but the high upfront cost may still be a barrier for some households.
- Weather Dependency: Solar power generation relies on sunlight. Cloudy days, storms, or extended periods of low sunlight, such as winter, can reduce system performance, making it less reliable without sufficient battery storage.
- Space Requirements: Installation requires adequate roof and or ground space for solar panels, which may not be feasible for all homes, especially in densely populated areas
- Energy Storage Limits: Battery capacity restricts how much energy can be stored and used during non-sunlight hours. Once the battery is depleted, the system cannot provide power and must be supplemented with grid electricity.
- Scalability vs Cost: While the system can be expanded to provide lighting to more areas, expanding will incur additional cost to which could become a limitation.
- Efficiency Loss: Energy losses occur during solar conversion, battery charging/discharging, as well as system operations, which will reduce overall efficiency.

5.3. Alternatives

In evaluating the proposed solar home lighting system, there are several alternatives to consider, each with its own set of tradeoffs compared to the proposed system:

- Conventional Grid-Based Lighting: Traditional and inexpensive, but offers no sustainability or backup during outages.
- Solar Lighting Without App Control: A solar home system with panels, battery, and LED setup with only physical switches. This alternative is more affordable but lacks monitoring, scalability, and smart functionality.
- Hybrid Solar System: Combines solar with a generator or grid connection for backup. This increases reliability but adds cost, complexity.
- Commercial Smart Home Lighting: Uses grid electricity to power smart LEDs with wifi control. These, however, don't reduce energy cost or environmental impact.

5.4. Impact

- Environmental Impact: Reduces carbon emissions by using solar energy and energy-efficient LEDs. There will still be some environmental footprint through battery and panel production.
- Societal Impact: Provides reliable lighting, improves home security, and enhances user convenience.
- Ethics: May raise concerns regarding data privacy on the mobile app. Affordability for lower-income households is also a factor, as well as the sourcing of materials used in the system.

Solar Home Lighting 3

Jack Holdridge
Kyle Walcher
Matthew Greer
Adam Garcia

FUNCTIONAL SYSTEM REQUIREMENTS

REVISION – 1
2 October 2025

Functional System Requirements

for

Solar Home Lighting

Prepared by:

Author Date

Approved by:

Jack Holdridge Date

John Lusher, P.E. Date

Hailey Ward Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	10/4/2025	Solar Home Lighting System		Draft Release

Table of Contents

List of Tables 4
List of Figures 5
1. Introduction 5
1.1. Purpose and Scope 5
1.2. Responsibility and Change Authority 5
2. Applicable and Reference Documents 6
2.1. Applicable Documents 6
2.2. Reference Documents 6
2.3. Order of Precedence 8
3. Requirements 8
3.1. System Definition 8
3.2. Characteristics 9
3.2.1. Functional / Performance Requirements 9
3.2.3. Electrical Characteristics 10
3.2.4. Environmental Requirements 12
3.2.5. Failure Propagation 12
4. Support Requirements 13
Appendix A: Acronyms and Abbreviations 14
Appendix B: Definition of Terms 15

List of Tables

No table of figures entries found.

List of Figures

Figure 1. Your Project Conceptual Image **1**
Figure 2. Block Diagram of System **4**

1. Introduction

1.1. Purpose and Scope

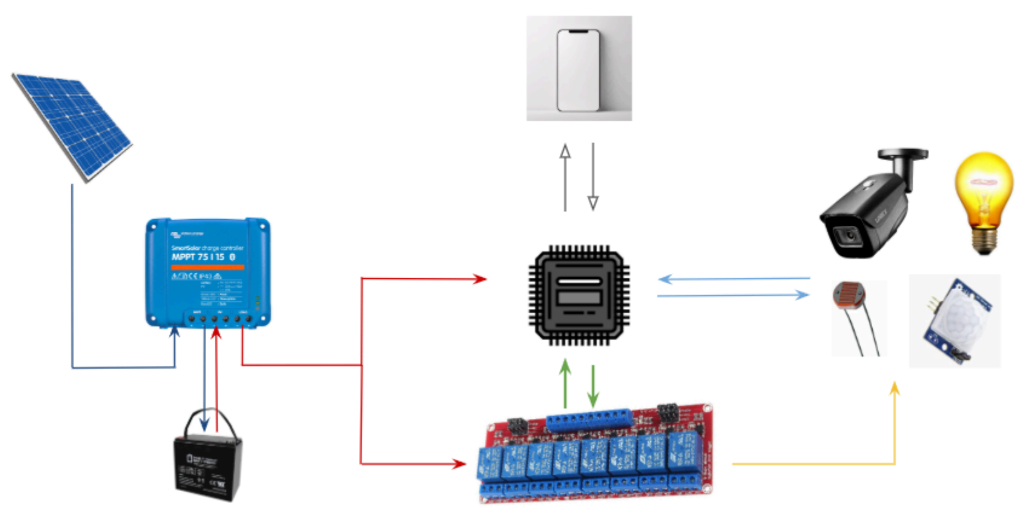
The purpose of this project is to design, develop, and demonstrate a fully integrated solar-powered residential lighting and security system capable of operating independently from the electrical grid. The system shall support both interior and exterior applications and shall combine the functionality of traditional lighting control with modern smart security features into a unified, renewable energy powered platform.

This system encompasses the complete development of a modular system architecture, including solar energy generation, battery storage, sensor driven automation, real time monitoring, and user control via both a local interface and mobile application. The system is intended to enhance residential security while reducing energy consumption and utility cost by leveraging solar energy as the primary power source.

For demonstration purposes, a function prototype will be built simulating a typical porch and foyer installation. However, the system is designed to be scalable and expandable to support multiple rooms and outdoor zones across a full residential property. The final implementation shall include motion detection, door sensors, environmental light sensing, programmable timers, and a network of lighting and surveillance devices. All of which will be managed through a central microcontroller and ESP32 based wireless communication.

The scope of this work includes hardware and software integration, embedded system development, sensor interfacing, application design, and functional validation of system performance under typical residential operating conditions.

Figure 1. Your Project Conceptual Image



The Solar Home Lighting System uses sunlight to power lights and small devices in a home. During the day, the solar panel will collect energy from the sun and send it to a battery for storage. The charge controller makes sure the battery charges safely and efficiently. When the sun goes down, the system uses the energy stored in the battery. It also has load control, which means it can decide which lights or devices get power first, depending on how much energy is available.

Users can connect to the system using a mobile app, which shows how much energy is being made, how full the battery is, and lets them turn lights on or off remotely. All the system's data, like how much power was used or if there were any problems, is saved in a database.

1.2. Responsibility and Change Authority

All project requirements shall be verified under the authority of the team captain, Jack Holdridge, who shall ensure each requirement is met in accordance with the project specification. No requirement may be altered without prior approval from both the team captain and the project sponsor, Professor Won Jang.

Subsystem Ownership is as follows:

- MPPT Charge Controller – Adam Garcia
- Physical Controls – Jack Holdridge
- IOS Application and Camera – Kyle Walcher
- MCU/Sensors & Database – Matthew Greer

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

Document Number / Title	Revision / Release Date	Description
Lighting Global SHS Quality Standards v2.5	Dec 2018	Defines minimum quality, durability, and safety for solar home system kits up to 350W. Includes testing protocols and eligibility criteria.
IEEE 1547a	2020	*IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces--Amendment 1: To*

		Provide More Flexibility for Adoption of Abnormal Operating Performance Category III
NEC Article 690	2023	*National Electrical Code for Solar Photovoltaic Systems. Covers wiring, grounding, and safety for PV installations.*
UL 1741	2021	*Standard for Inverters, Converters, Controllers, and Interconnection System Equipment for Use With Distributed Energy Resources.*
ANSI C119.6	2024	*Electric Connectors—Non-Sealed, Multiport Connector Systems Rated 600 V or Less for Aluminum and Copper Conductors.*
IEC 62852	2014	*Connectors for DC-application in photovoltaic systems - Safety requirements and tests*
NEC Article 480	2023	*National Electrical Code for Storage Batteries*

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

Document Number / Title	Revision / Release Date	Description
ESP32 TRM (Version 5.5)	2025.08	*ESP32 - Technical Reference Manual Version 5.5*
iOS Human Interface Guidelines	2025	*Apple Developer Guidelines for iOS Application Design*
ISO 9241-210	2019	*Ergonomics of human-system interaction*
ISO/IEC 14882	2024	*Programming languages — C++*

IEEE 802.11	2020	*IEEE Standard for Information Technology--Telecommunications and Information Exchange between Systems*
ISO/IEC 9075-1	2023	*Information technology — Database languages SQL*

2.3. Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings or other documents that are invoked as “applicable” in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICDs that have their relevant documents considered to be incorporated as cited.

3. Requirements

3.1. System Definition

The Smart Solar Residential Lighting and Security System integrates four main subsystems, each fulfilling a critical role in delivering a reliable, energy efficient solution for residential applications. The solar power network centers on an MPPT charge controller that optimizes energy harvesting by regulating the variable DC output from the solar panels to efficiently charge the battery bank. This controller communicates with the battery management systems and central microcontroller to ensure continuous power availability.

The physical control subsystems provide a user-friendly touch screen interface for lighting control and essential system status monitoring. It also hosts DC voltage conversions and serves as the central connection point for all system components, ensuring efficient power distribution and safe operation compliant with residential electric codes.

The lighting and security monitoring functions are supported by an iOS application paired with an ESP32-EYE camera module. This subsystem streams live video feeds and uses AI-powered motion detection to alert users of security events, while securely logging data to a cloud database for later review.

At the core of the system, the microcontroller and sensor subsystem collects environmental and motion data, executes control logic, and maintains real-time communication with the mobile app and cloud database via Wi-Fi. Sensor inputs trigger lighting adjustment and security alerts, ensuring responsive and adaptive system behavior.
Together these interconnected subsystems create a cohesive, scalable solution that provides solar-powered, automated lighting and security, addressing the need for sustainable energy use and enhancing residential awareness.

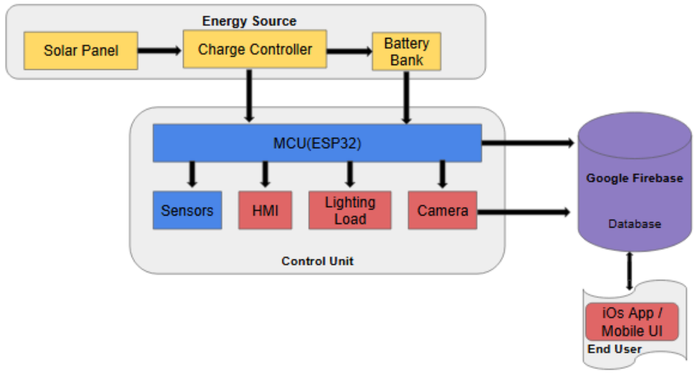


Figure 2. Block Diagram of System

3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Lighting Duration

The system shall provide continuous home/residential operation of exterior and interior lighting for a minimum of eight hours during night house conditions, powered solely from solar powered energy.

Rationale: This ensures that residential homes can receive reliable illumination throughout the whole night that is not dependent on grid electricity. This is critical to both the usability of the system and ability to reduce monthly costs.

3.2.1.2. Motion Detection Accuracy

The system will be able to detect and log human motion events with an accuracy of at least 90% under typical residential operating conditions, including varying lighting level and regular environmental noise

Rationale: Reliable detection of motion is essential for the systems role as a resource for security. A reliable, High detection accuracy will minimize false negatives and ensures timely push notifications that are generated for the homeowner

3.2.1.3. Camera Streaming Latency

The mobile iOS application developed in SwiftUI shall stream live video from the ESP32-EYE camera with no more than two seconds when operating over standard WiFi

Rationale: The requirement will ensure that the homeowner experiences video feedback that is as close to real time as possible for responsive decision making for the user

3.2.1.4. Database Storage

The system shall maintain at least thirty days of stored even snapshot data in its cloud (firebase) for later review and analysis

Rationale: A month of storage strikes between useful historical/security data and practical storage constraints, allowing homeowners to review the past without over the top resource costs.

3.2.1.5. User Control and Responsiveness

The iOS application will allow the user to toggle lighting states (ON / OFF) and received push notifications of change within 2 to 3 seconds of command execution

Rationale: Ensures the mobile will feel seamless and responsive to the user, which is essential for modern usability

3.2.1.6. Motion Triggered Lighting

The system shall activate exterior lighting once triggered by motion detection for at least 30 seconds unless reset by further activity

Rationale: Automatic light response provides both security and deterrence. This will provide users more safety and controllability within the application.

3.2.2. Physical Characteristics

3.2.2.1. Subsystem Size

The physical control unit shall not exceed 15 x 20 cm footprint, while individual sensors and camera shall fit within a 10x10x10 cm volume.

Rationale: Ensures compact sizing for ease of installation in tight residential spaces while maintaining discreet placement for cameras and sensors

3.2.2.2. Battery Housing

The battery housing shall be fully enclosed in a weather resistant casing that is also ventilated

Rationale: Protects deep cycle battery from weather exposure while allowing airflow

3.2.2.3. Cable Management and Mounting

All interconnections between the subsystems shall utilize standard cable runs and shall be designed for mounting with basic tools

Rationale: Simplifies installation and ensures compliance with safe residential practices

3.2.3. Electrical Characteristics

3.2.3.1. Inputs

a. The Solar Home Lighting System shall tolerate any combination and sequence of input signals as defined in the Interface Control Document (ICD) without sustaining damage, degradation of life expectancy, or operational malfunction. This resilience applies equally whether the system is energized or de-energized.

b. No permissible sequence of input commands shall induce system failure, reduce expected service life, or impair functionality.

Rationale: The system is engineered to minimize the risk of damage or malfunction arising from user or technician actions, ensuring reliability in real-world operation.

3.2.3.1.1. Power Consumption

a. The maximum peak power of the system, including operation of lights, camera, sensors, wireless communication, shall not exceed 50 watts

Rationale: Limiting peak power draw ensures easy compatibility with mid level solar generation capacity and affordable battery storage. This will enable the system to remain cost effective while still meetings performance requirements

3.2.3.1.2. Input Voltage Range

The system shall accept DC input from 12 to 24 volts, compatible with standard solar batteries

Rationale: Flexibility will allow for seamless integration with commercially available solar panels and storage systems without custom hardware

3.2.3.1.3 DC/DC Conversion Efficiency

All DC/DC converters (bucks and boost) shall maintain a minimum conversion efficiency of 85% under load conditions

Rationale: High efficiency is required to minimize energy loss and maximize usable solar energy

3.2.3.1.4. Overcurrent and Short Circuit Protection

The system shall include overcurrent protection on all major supply rails and shall withstand short circuit episodes without permanent damage to the system

Rationale: Protects the system from wiring faults, load failures, and user error. This will ensure safety and reliability

3.2.3.1.4. Data and Control Signals
All MCU and sensor interfaces should conform to logic levels of 3.3 V with proper protection for I/O pins

Rationale: Aligns with ESP32 electrical limits, ensuring proper and safe communication between subsystems

3.2.3.2. Outputs

3.2.3.2.1. Energy Output
The Solar Home Lighting System shall provide a regulated DC output compatible with 12V appliances.

Rationale: Ensures LED lighting can be used off-grid.

3.2.3.2.2. Battery Status Output
The Solar Home Lighting System shall include a digital interface to provide the battery charge level.

Rationale: The user and technician can monitor battery performance and plan accordingly.

3.2.3.2.3. Raw Video Output
The Solar Home Lighting System application shall include a raw video interface to support external recording.

Rationale: Too much data to store videos; only photos are stored internally. It would be used for security.

3.2.3.2.4. Connectors
The Solar Home Lighting System shall use weatherproof DC connectors compliant with IEC 62852.

Rationale: Ensures safe connections in outdoor environments.

3.2.3.2.5. Wiring
The Search and Rescue System shall follow the guidelines outlined in NEC Article 690 and use UL-listed cables.

Rationale: Ensures electrical safety and minimizes fire risk.

3.2.4. Environmental Requirements

The Solar Home Lighting System shall be designed to withstand and operate in the environmental conditions and laboratory tests specified in the following subsections.

Rationale: These requirements are based on scenarios where the system must maintain reliable operation under varying and harsh weather.

3.2.4.1. Thermal

The Solar Home Lighting System shall operate between –20°C to +55°C.

Rationale: This range accommodates extreme hot and cold conditions in the U.S.

3.2.4.2. Rain

The Solar Home Lighting System shall withstand heavy rainfall and storm conditions, including exposure to high precipitation rates.

Rationale: Ensures waterproofing and short circuits, and corrosion.

3.2.4.3. Solar Radiation

All components exposed to outdoor elements shall be UV-resistant and capable of withstanding high solar irradiance.

Rationale: Prolonged exposure to sunlight can degrade plastics and coatings of components.

3.2.4.2. Wind

The Solar Home Lighting System shall withstand 150 km/h wind speeds without structural failure and detachment.

Rationale: High winds pose a risk of damage to rooftop installations.

3.2.5. Failure Propagation

The Solar Home Lighting System shall not allow propagation of faults beyond its system interface, ensuring that failures within the system do not affect connected external devices or the grid interface.

3.2.5.1. Failure Detection, Isolation, and Recovery (FDIR)

3.2.5.1.1 Built-In Test (BIT)

The Solar Home Lighting System shall include an internal subsystem that generates diagnostic signals and evaluates system responses to determine the presence of faults.

3.2.5.1.1.1. BIT Critical Fault Detection

The BIT shall detect critical faults in the Solar Home Lighting System with a minimum detection rate of 95 percent.

Rationale: This requirement ensures reliable fault detection in remote or off-grid environments where manual diagnostics are impractical.

3.2.5.1.1.2. BIT False Alarms

The BIT shall have a false alarm rate of less than 5 percent.

Rationale: Minimizing false alarms reduces unnecessary maintenance interventions and improves user trust in system diagnostics.

3.2.5.1.1.3. BIT Log

The BIT shall record the results of each diagnostic test in a log stored within the Solar Home Lighting System. This log shall be accessible for retrieval and clearing by maintenance personnel.

Rationale: Persistent logging supports fault analysis and long-term reliability tracking.

3.2.5.1.2 Isolation and Recovery

The Solar Home Lighting System shall support fault isolation and recovery by enabling subsystems to be reset, disabled, or reconfigured based on BIT results.

Rationale: This capability ensures that faults can be contained and mitigated without requiring full system shutdown, preserving partial functionality in critical lighting scenarios.

4. Support Requirements

The Smart Solar Residential Lighting and Security System requires stable Wi-Fi connection within the home to enable communication between the control panel and mobile applications. Users must provide access to Wi-Fi and maintain the solar panel installation in an area with sufficient sunlight for optimal performance. A compatible iOS device is additionally required to operate the mobile application and receive security notifications.

The system package includes all necessary hardware components, such as solar panels, batteries, control boards, sensors, and lighting units. Users will use the iOS mobile application for wireless control and monitoring. A comprehensive user manual and an in-app walkthrough guide are provided to assist with installation, operation, and troubleshooting.

Technical support is available via email and phone during standard business hours. In the event of hardware malfunction or operational issues in the field, users can contact support for remote diagnostic and troubleshooting. Field issues related to solar panel position or battery performance may require an on-site inspection or user intervention as guided by support. Periodic software updates will be pushed automatically to enhance system functionality and security.

Appendix A: Acronyms and Abbreviations

MPPT	Maximum Power Point Tracking
PWM	Pulse Width Modulation
Wi-Fi	Wireless Fidelity
iOS	iPhone Operating System
AC	Alternating Current
DC	Direct Current
RMS	Root Mean Square
HMI	Human Machine Interface
ICD	Interface Control Document
MCU	Microcontroller Unit
km/h	Kilowatts per hour
LED	Light Emitting Diode

Appendix B: Definition of Terms

This appendix provides definitions for specialized technical terms and acronyms used throughout the Functional System Requirements document. These definitions ensure consistency and clarity when interpreting the system description, requirements, and validation criteria.

Solar Irradiance	The amount of solar power received per unit area, measured in watts per square meter (W/m^2).
MPPT (Maximum Power Point Tracking)	A control method used by the charge controller to continuously adjust voltage and current for maximum solar-panel power output under varying sunlight conditions.
Lux	A unit of illuminance equal to one lumen per square meter, used to quantify visible light intensity detected by the system's ambient-light sensor.
HMI (Human-Machine Interface)	The touchscreen display and control interface that allows users to view system data and issue lighting or configuration commands.
Firebase	A cloud-based database service used to store sensor data, camera images, and system status information for mobile-app access.
ESP32	A low-power microcontroller with integrated Wi-Fi and Bluetooth, used as the system's primary processing and communication unit.

Solar Home Lighting 3

Jack Holdridge
Kyle Walcher
Matthew Greer
Adam Garcia

Interface Control Document

REVISION – 1
25 January 2018

Interface Control Document
for
Solar Home Lighting

Prepared by:

Author Date

Approved by:

Jack Holdridge Date

John Lusher II, P.E. Date

T/A Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	10/4/2025	Solar Home Lighting System		Draft Release

Table of Contents

No table of figures entries found....5

1. Overview....5

2. References and Definitions....6

- 2.1. References....6
- 2.2. Definitions....6

3. Physical Interface....8

- 3.1. Weight....8
- 3.2. Dimensions....8
  - 1.1.1. Dimension of subsystem name....8
- 3.3. Mounting Locations....8

4. Thermal Interface....9

5. Electrical Interface....10

- 5.1. Primary Input Power....10
- 5.2. Polarity Reversal....10
- 5.3. Signal Interfaces....10
- 5.4. Video Interfaces....10
- 5.5. User Control Interface....10

6. Communications / Device Interface Protocols....11

- 6.1. Wireless Communications (WiFi)....11
- 6.2. Host Device....11
- 6.3. Video Interface....11
- 6.4. Device Peripheral Interface....11

List of Tables

No table of figures entries found.

List of Figures

No table of figures entries found.

1. Overview

This Interface Control Document (ICD) defines how the subsystems of the Smart Solar Home Lighting System will interface with each other. It covers the connections between the solar power, battery, sensors, microcontroller, lighting, camera module, and mobile applications. The ICD outlines electrical, data, and communication interfaces to ensure reliable system integration and operation.

2. References and Definitions

2.1. References

Lighting Global SHS Quality Standards v2.5
Dec 2018

IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces (IEEE 1547a-2020)
2020

National Electrical Code for Solar Photovoltaic Systems (NEC Article 690)
2023

Standard for Inverters, Converters, Controllers, and Interconnection System Equipment for Use With Distributed Energy Resources (UL 1741)
2021

ANSI Electric Connectors (ANSI C119.6)
2024

Storage Batteries (NEC Article 480)
2023

2.2. Definitions

MPPT	Maximum Power Point Tracking
PWM	Pulse Width Modulation
Wi-Fi	Wireless Fidelity
iOS	iPhone Operating System
AC	Alternating Current

DC	Direct Current
RMS	Root Mean Square
HMI	Human Machine Interface
ICD	Interface Control Document
MCU	Microcontroller Unit
kWh	Kilowatts-hour
LED	Light Emitting Diode
USB	Universal Serial Bus
UART	Universal Asynchronous Receiver-Transmitter
BLE	Bluetooth Low Energy
LCD	Liquid Crystal Display
PIR	Passive Infrared Sensor
HTTP	Hypertext Transfer Protocol
MJPEG	Motion JPEG
VGA	Video Graphics Array
IEEE	Institute of Electrical and Electronics Engineers

3. Physical Interface

3.1. Weight

3.1.1. Physical Controls Subsystem

Component	**Weight (g)**	**Number of Items**	**Total Weight (g)**
ESP32	31	1	31
2.8” TFT LCD	40	1	40
6 Channel Relay Board	90	1	90
PIR Motion Sensor	10	1	10
Reed Switch Door Sensor	10	1	10
Daylight Sensor	17	1	17
LED Recessed Ceiling Light	9.5	4	38
LED Flood Security Light	296	2	592
Total Weight (Approx.)			**~828 grams**

3.1.2. ESP32 Camera and Application Subsystem

Component	Weight (g)	Number of items	Total Weight (g)
Speed Studio XIAO ESP32 S3 Sense	6.6	1	6.6
Total Weight (Approx.)			**~6.6 grams**

3.1.3 MPPT Charge Controller

Component	Weight (g)	Number of Items	Total Weight (g)
200W Solar Panel	7520	1	7520
12V Lead Acid Battery	10433	1	10433
ESP32	31	1	31
Total Weight (Approx.)			**~17,984 grams**

3.2. Dimensions

3.2.1. Physical Controls Subsystem

Component	Length (in)	Width (in)	Height (in)
ESP32	2.05	1.18	0.59
2.8” TFT LCD	2.8	2	0.3
6 Channel Relay Board	7.17	4.49	0.75
PIR Motion Sensor	1.5	1.5	0.75

Daylight Sensor	2.06	3.0	0.25
LED Recessed Ceiling Light	2.75	2.75	0.62
LED Flood Security Light	4.53	2.76	3.35

3.1.2. ESP32 Camera and Application Subsystem

Component	**Length (in)**	**Width (in)**	**Height (in)**
Speed Studio XIAO ESP32 S3 Sense	0.83	0.70	0.59

3.1.3 MPPT Charge Controller

Component	**Length (in)**	**Width (in)**	**Height(in)**
200W Solar Panel	42.12	0.38	25.24
12V Lead Acid Battery	6.5	5	7.06
ESP32	2.05	1.18	0.59

3.3. Mounting Locations

3.3.1. Physical Controls Subsystem Panel

The subsystem is intended for indoor mounting near the main residential access point (foyer, hallway, or near a utility closet). Mounting option include:

- Wall mounting: using pre-drilled holes and drywall anchors
- Flush mounting: Inside low-voltage wall box or structured wiring panel

Mounting location must allow access to DC voltage lines and Wi-Fi coverage, and should be within visible and reachable distance for user interaction.

3.3.1. Physical Controls Subsystem Sensors

- **Motion Sensor**
    - The motion sensor will be mounted below the ESP32-EYE camera, allowing the 30 feet of motion detection range to be utilized for camera triggering purposes.
- **Reed Switch Door Sensor**
    - This will be mounted at the top of the door frame to indicate door status.
- **Daylight Sensor**

  - This will be mounted to the roof alongside the solar panel to get an accurate light level readout.

3.3.2. ESP32 Camera and Application Subsystem

- **ESP32-Camera**
  - Indoor facing outwards in wall mounted casing. In outdoor settings, the esp32-camera will be placed on a wall mount in weather-resistant housing with transparent lens cover.

3.1.3 MPPT Charge Controller

- **12V battery and charge controller**
  - Must be in proximity to one another, preferably in the same location.
  - Indoors or safe from harsh weather elements, like a cabinet or storage house.
- **Solar Panel**
  - Must be on an elevated outdoor structure with full sun exposure.
  - Securely mounted for windy conditions.

4. Thermal Interface

4.1.1. Physical Controls Subsystem

The Physical Controls subsystem generates minimal thermal load during operation. Passive cooling through ventilation slots in the enclosure is sufficient under standard indoor ambient temperatures.

- Cooling method: passive ventilation
- Heat dissipation components: ESP32 controller, voltage regulators
- Thermal Management: small aluminum heatsinks will be affixed to the heat dissipating components
- No active cooling interface is required

The enclosure material is ABS plastic, suitable for residential installations. The operating temperature range is between 0-70°C.

4.1.2. ESP32 Camera and Application Subsystem

Typical operating temperature for a Speed Studio XIAO ESP32 S3 Sense is between 32°F – 122°F (0°C – 50°C). The heat dissipation is minimal (<1 W power draw) but airflow will be utilized for enclosed environments

4.1.3. MPPT Charge Controller Subsystem

No active cooling is required as passive cooling is sufficient under standard indoor ambient conditions, with heatsinking applied to MOSFETs and inductors. Slotted ventilation can assist in cooling. Thermal isolation is maintained between heat-generating components and temperature-sensitive sensors.

5. Electrical Interface

5.1. Primary Input Power

5.1.1. Physical Controls Subsystem

The physical control will be powered by the DC battery bank from the charge controller subsystem.

- Input Voltage Range: +11.5 VDC to +13.0 VDC (nominal)
- Input Type: 2-pin terminal block (fused type)

5.1.2. ESP32 Camera and Application Subsystem

Supply voltage will be 3.3 V regulated via USB-C or battery pack. Typical current draw will be 160-200 mA during streaming, with peak current up to 500 mA.

4.1.3. MPPT Charge Controller Subsystem

The primary input power for both the battery and the charge controller will be directly from the Solar Panel.

5.2. Voltage and Current Ratings

5.2.1. Physical Controls Subsystem Internal Power Consumption

Component	**Input Voltage (VDC)**	**Current (mA)**	**Power (W)**
ESP32 Microcontroller	3.3	250	0.825
Touchscreen LCD	3.3	150	0.495
Relay Control Board	3.3	150	0.495
Voltage Regulator (3.3V)	12	150	1.8

PIR Motion Sensor	3.3	3	0.009
Reed Switch Door Sensor	3.3	<1	N/A
Daylight Sensor	3.3	2	0.007
Internal Subtotal			**~3.7 W**

5.2.2. Physical Controls Subsystem Lighting Output Load

Load Type	**Quantity**	**Power per Unit (W)**	**Total Load (W)**
LED Recessed Ceiling Light	4	2.5	10
LED Flood Security Light	2	10	20
Total Peak Load			**30 W**

5.2.2. Camera Subsystem and Application

Component	Input Voltage	Typical Current	Peak Current	Power
XIAO ESP32-S3	3.3 V	180 mA	300 mA	0.59 W

3.1.3 MPPT Charge Controller

Load Type	**Input Voltage (VDC)**	**Current (mA)**	**Power (W)**
200W Solar Panel	18.8	10.6	200
12V Lead Acid Battery	~14V	100 mAh	
ESP32	3.3	250	0.825
Total Peak Load			**~201 W**

5.3. Polarity Reversal

To protect against reverse polarity connection on the 12VDC input, the physical control subsystem includes an in-line Schottky diode for polarity protections. This diode will be rated for a current exceeding the max peak load continuous forward current, providing reliable protection for the subsystems peak load. Under normal load conditions the diode permits current flow with a typical voltage drop of approximately 0.5V, resulting in minimal power loss.

5.4. Signal Interfaces

5.4.1. Physical Controls Subsystem Input and Output Signals

- Digital Inputs:
  - Reed switch door sensors
  - PIR motion sensors
  - I2C capacitive touch screen

- Analog Inputs:
  - Daylight Sensor (LDR voltage divider)

- Outputs:
  - 6 relay outputs for 12VDC lighting control

4.1.3. MPPT Charge Controller Subsystem

UART (Serial)	Communication between MCUs	ESP32 ↔ Battery Management System
ADC (Analog Input)	Reads analog voltages from sensors	ESP32 ↔ Solar panel voltage, current sensors
PWM (Pulse Width Modulation)	Controls power stages	ESP32 ↔ Buck Converter, LED brightness control
Wi-Fi/BLE	Wireless communication with the app	ESP32 ↔ iOS App, Firebase Database

5.6. User Control Interface

5.6.1. Physical Controls Subsystem Control Screen

Interface: 2.8” I2C Capacitive Touch screen

- Functions:
  - Toggle lights ON/OFF

- View battery charge statistic
- View system operating time capability

6. Communications / Device Interface Protocols

6.1. Wireless Communications (WiFi)

Standard: IEEE 80.2.11 b/g/n at 2.4 GHz
Function: Provides primary communication path between the ESP32 camera subsystem and iOS mobile application
Bandwidth: MJPEG video stream operates at up to 2 Mbps depending on resolution. MQTT sensor/control messages operate under 10 kbps, which ensures minimal overhead

6.2. Host Device

Mobile Device: iOS smartphone or tablet running on Flutter/SwiftUI based application.

6.3. Video Interface

Type: MJPEG over HTTP streaming from an onboard ESP32 server
Protocol: HTTP GET request handled by ESP32 web server library
Resolution: VGA (640x480) 15 fps

6.4. Device Peripheral Interface

Maybe a serial port (UART) with a custom protocol, in which case you describe the protocol in detail.

Solar Home Lighting 3

Jack Holdridge
Kyle Walcher
Matthew Greer
Adam Garcia

Schedule and Validation

REVISION – Draft
4 October 2025

Schedule:

Work	End Date	Owner	Status	Date Completed
Concept of Operations	**9/17/2025**	All	Complete	9/17/2025
Installed ESP-IDF, created Firebase database	**9/25/2025**	Matthew Greer	Complete	9/25/2025
Established firmware framework & Wi-Fi connection test	**9/30/2025**	Matthew Greer	Complete	9/30/2025
Functional System Requirements	**10/4/2025**	All	In Progress	—
Interface Control Document	**10/4/2025**	All	In Progress	—
Designed DC converters, touchscreen connection, PCB layout	**10/6/2025**	Jack Holdridge	Planned	—
Designed and simulated MPPT circuit, PCB, and other components	**10/6/2025**	Adam Garcia	Planned	—
Begin coding, implement initial control logic, perform calibration test	**10/6/2025**	Matthew Greer	In Progress	—
Progress Update 1	**10/6/2025**	All	Planned	—
Test connection between ESP32-EYE and iOS application	**10/10/2025**	Kyle Walcher	Planned	—
Receive circuit components & program touch screen interface	**10/20/2025**	Jack Holdridge	Planned	—
Implement MCU-to-Firebase communication & validate sensors	**10/18/2025**	Matthew Greer	Planned	—

Validate sensor accuracy (lux, PIR, temp)	**10/19/2025**	Matthew Greer	Planned	—
Midterm Presentation	**10/20/2025**	All	Planned	—
Progress Update 2	**10/20/2025**	All	Planned	—
Test converters & MCU program	**10/20/2025**	Adam Garcia	Planned	—
Camera fully embedded with polished UI	**10/20/2025**	Kyle Walcher	Planned	—
Circuit PCB, MCU, and physical controls integration	**11/3/2025**	Adam Garcia, Jack Holdridge	Planned	—
Assemble PCB and connect sensors, load, and HMI	**11/3/2025**	Jack Holdridge	Planned	—
Full integration: Sensors → MCU → Database → App, with power tests	**11/3/2025**	Matthew Greer	Planned	—
Test functionality with OpenAI + weather data	**11/3/2025**	Kyle Welcher	Planned	—
Progress Update 3	**11/6/2025**	All	Planned	—
Validate communication across subsystems and finalize control logic	**11/17/2025**	Matthew Greer	Planned	—
Validate battery & loads are safely powered	**11/17/2025**	Adam Garcia	Planned	—
Validate assembled HMI functions as expected	**11/17/2025**	Jack Holdridge	Planned	—
Validate application communicates seamlessly with camera and database	**11/17/2025**	Kyle Walcher	Planned	—
Progress Update 4	**11/17/2025**	All	Planned	—
Subsystems Demonstration	**12/3/2025**	All	Pending	—
Final Report	**12/6/2025**	All	Pending	—

Validation Plan:

Task	Specification	Result	Owner
Verify MPPT charge controller performance	Charge controller maintains >90% efficiency under varying solar input	Pass/Fail – measure voltage/current with simulated solar load	Adam Garcia
Battery/load safety validation	No over-current or undervoltage during charging/discharging	Pass/Fail – multimeter + load test	Adam Garcia
Measure peak internal power consumption	≤ 5W total internal power	Pass/fail - multimeter + load test	Jack Holdridge
Physical controls HMI functionality	Touchscreen inputs register with <500 ms latency and <5% error	Pass/Fail – 50 button press trials	Jack Holdridge
Verify data update on HMI screen	System status (light on/off, battery charge information)	Pass/Fail - 50 test updating information parameters	Jack Holdridge
Sensor detection (PIR, lux, temperature)	PIR detects motion ≥6 m; Lux sensor ±5% vs. calibrated meter; Temp ±2 °C	Pass/Fail – log sensor readings vs. reference instruments	Matthew Greer
Control logic (auto dusk/dawn)	Lights turn on at ≤20 lux, off at ≥40 lux (5 lux hysteresis)	Pass/Fail – dimmable light sweeps	Matthew Greer
Measure lighting output load	Total peak lighting load: 30W	Pass/Fail - multimeter	Jack Holdridge
MCU-to-Database communication	Write latency ≤1.0 s median (95% < 2.0 s), ≥99% successful writes	Pass/Fail – 200 test messages logged	Matthew Greer
App-to-Database refresh	UI updates ≤5 s after DB change; seamless	Pass/Fail – stopwatch across 50	Kyle Walcher

	camera integration	updates	
Camera integration	ESP32-EYE streams still frames into app reliably at ≥90% success	Pass/Fail – 100 test captures, record drop rate	Kyle Walcher
Power consumption check	System average current supports ≥12 h continuous operation at average load draw	Pass/Fail – 12 h load run logged with probe	Adam Garcia + Jack Holdridge
Integration test (end-to-end)	Sensor → MCU → DB → App → Lighting works as expected	Pass/Fail – 20 functional cycles observed	All
System reliability	Continuous operation for 7 days with ≤1 reset/fault	Pass/Fail – uptime logs, watchdog triggers	All
Environmental tolerance	Operates at 0–40 °C, 20–90% RH	Pass/Fail – lab stress tests	All